

PROBLEM STATEMENT

Infodemic Containment

Misinformation Spreaders vs. Fact-Checking Moderators on a Social Network

Type: Multi-agent adversarial simulation | Agent types: 2

1. Background and Motivation

Social platforms let a claim reach thousands of people within hours of being posted. Empirical work on Twitter (Vosoughi, Roy & Aral, *Science*, 2018) found that false stories diffuse farther and faster than true ones, largely because they are amplified through a small number of well-connected accounts.

Verification does not scale the same way. Fact-checking is slow, evidence-dependent, and limited by reviewer capacity, so moderators can only check a small fraction of flagged content in any given window. Aggressive takedowns carry their own cost: wrongly suppressing true content damages user trust and platform credibility.

The result is a race between propagation speed and verification capacity, played out over a network where influence is very unevenly distributed.

2. Problem Definition

Given a social network, a stream of claims (some true, some false, with labels hidden from moderators), and a fixed per-round verification budget, **how should a moderator decide which claims to verify, when, and what action to take, in order to minimize total misinformation exposure while limiting wrongful takedowns, against an adversary that actively adapts its spreading strategy?**

This is a sequential decision problem under uncertainty, with three difficulties built in:

- **Incomplete information.** Moderators never see ground truth. They infer it from claim content, propagation velocity, and source credibility history.
- **Scarce resources.** Only B claims can be checked per round, so every check has an opportunity cost.
- **An adaptive opponent.** The spreader observes moderator behavior and changes tactics, so a fixed policy gets exploited.

3. Why This Is an Agent Problem

A single-agent search or optimization framing fails because the environment contains another goal-directed agent whose actions change the outcome of yours. The setting is therefore a **multi-agent, mixed-motive environment with exactly two agent types**.

	Spreader Agent	Moderator Agent
Goal	Maximize exposed nodes before containment	Minimize total exposure + $\lambda \cdot$ false-positive takedowns
Actions	Post, reshare, create sockpuppets, target hubs	Flag, quarantine / rate-limit, publish counter-claim, escalate
Percepts	Local engagement, visible moderator activity	Claim content, propagation velocity, source credibility history
Constraint	Limited posts and accounts per round	Verification budget B per round, detection delay

Ordinary users are **not agents**. They are passive nodes in the environment that reshare probabilistically, which keeps the design at exactly two agent types.

4. Environment Specification

The environment is a scale-free graph $G = (V, E)$ of roughly 500 to 1000 nodes, chosen because real social networks have heavy-tailed degree distributions with a few very influential hubs.

- **Node states:** clean, exposed, quarantined
- **Edge weights:** reshare probability $p(u, v)$, modulated by a trust score
- **Claims:** synthetic predicates with hidden true/false labels, checkable against a knowledge base
- **Dynamics:** Independent Cascade. Each newly exposed node gets one chance to pass the claim to each neighbor with probability $p(u, v)$

Environment property	Justification
Partially observable	Moderators infer truth from evidence; spreaders do not know the exact detection threshold
Stochastic	User resharing is probabilistic (Independent Cascade Model)
Multi-agent, mixed-motive	Adversarial goals, with neutral bystander nodes reacting probabilistically in between
Dynamic	The network state changes between an agent's turns
Sequential	Containment decisions now change what the spreader can do later

5. Decision Problems to Be Solved

Spreader

1. Where to seed the claim: a random node or a high-centrality hub?
2. When to deploy sockpuppets, and how to react to visible moderator activity?

Moderator

1. Which claims in the queue deserve the B checks this round? Ranking uses estimated impact (propagation velocity $\times$ reach of current carriers) rather than arrival order.
2. How to verify? Claims are represented as predicates and matched against a knowledge base using Horn-clause rule-based inference.
3. What to do when the knowledge base cannot resolve a claim? The options are to escalate, rate-limit provisionally, or ignore. This is where the limits of knowledge representation show up honestly.

6. Objectives and Trade-offs

- **Spreader:** maximize $E[\text{exposed nodes at horizon } T]$
- **Moderator:** minimize **Total Exposure** + $\lambda \cdot \text{FalsePositives}$

The parameter λ controls the moderator's caution. A low λ produces an aggressive moderator that contains rumors fast but wrongly suppresses true content. A high λ produces a cautious moderator that avoids collateral damage but lets rumors run longer. Sweeping λ and reporting the resulting exposure vs. false-positive frontier is a strong analytical result.

7. Stress Scenario: Coordinated Bot-Swarm Shock

At a scheduled round, many spreader instances activate simultaneously, mimicking a breaking-news panic. This tests whether the moderator's policy is robust or only works on the happy path. The system is expected to **re-prioritize under load** rather than continue with its pre-shock queue.

8. Research Questions and Hypotheses

ID	Statement
RQ1	How much does informed play beat uninformed play for each side?
H1	Centrality-targeted spreading reaches significantly more nodes than random targeting at equal post budget.
H2	Impact-weighted moderation beats FIFO and random moderation on total exposure at equal B.
RQ2	How does the system behave under the bot-swarm shock?
H3	An adaptive moderator recovers containment within k rounds, while a static policy does not.
RQ3	How does λ shape the exposure vs. false-positive trade-off?

9. Evaluation Plan

A 2×3 experiment matrix (six conditions), with multiple random seeds per condition since the process is stochastic:

Spreader \ Moderator	Random	FIFO	Impact-weighted
Random targeting	C1	C2	C3
Centrality-informed	C4	C5	C6

H1 is tested by comparing rows within a column (for example C1 vs. C4). H2 is tested by comparing columns within a row (for example C4 vs. C6). H3 is tested by applying the shock in each condition and comparing recovery. Sweeping λ on the moderator side addresses RQ3.

Metrics per run:

- Total exposure (area under the infection curve)
- Peak infection
- Time-to-containment
- False-positive rate
- Post-shock recovery time

10. Scope and Assumptions

- Claims are synthetic predicates, with no NLP or real-world content.
- The simulator holds ground truth. Moderators receive only noisy evidence, and this asymmetry is what makes the environment partially observable.
- Graph structure is static apart from node state changes and sockpuppet creation.
- The knowledge base is small and hand-authored, so results characterize the decision strategy, not real-world fact-checking accuracy.

11. Expected Contribution

The project shows quantitatively where informed search and resource-aware prioritization beat uninformed baselines in an adversarial, partially observable setting, and how a rule-based knowledge representation layer fits into a larger agent decision loop.